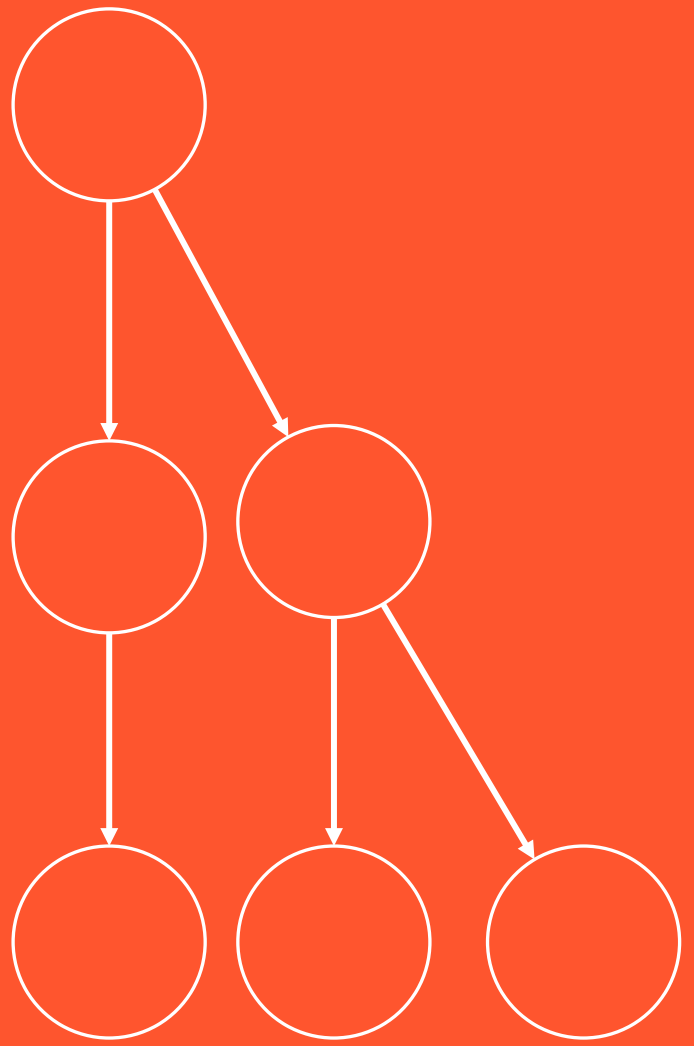


# BCOG 100

Perception and Action

Lecture 8.2 — Recovering a World from Moving Senses

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

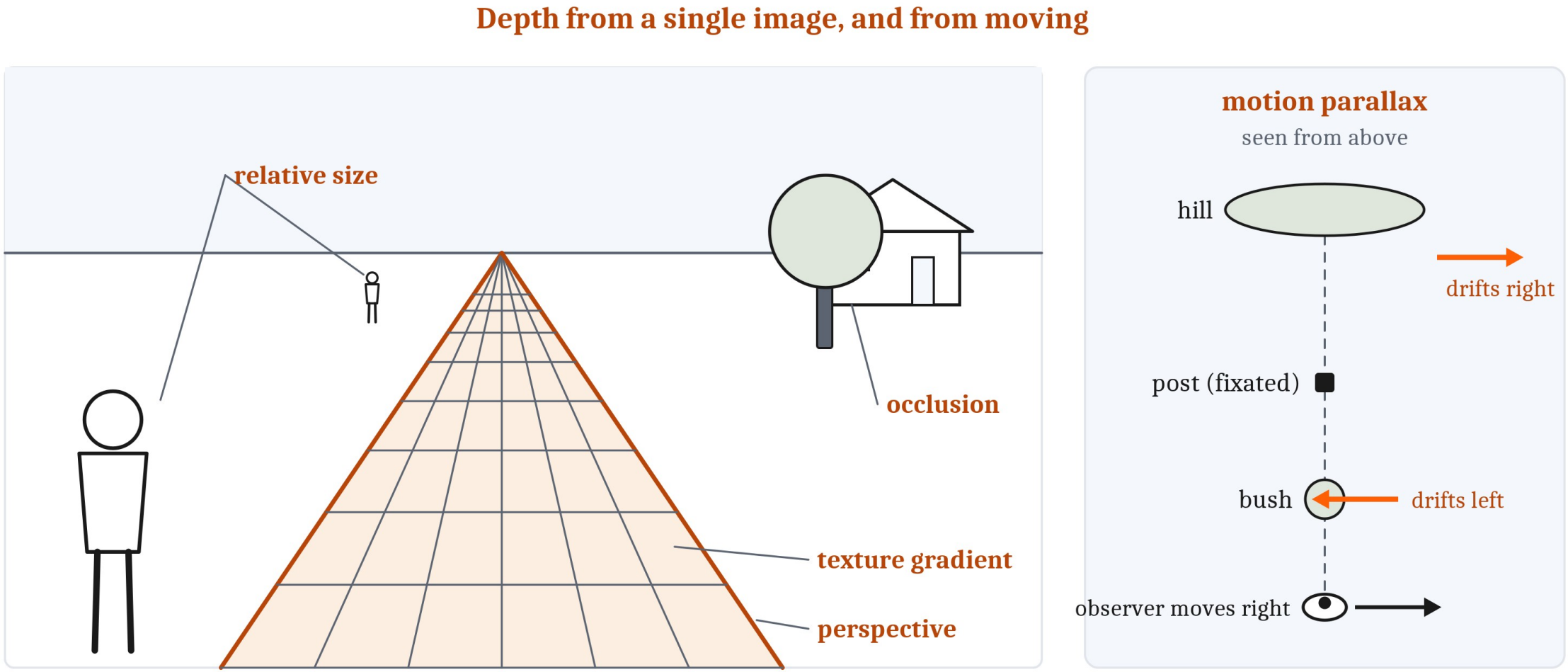
# To Start: Why Can No One Tickle Themselves?

**In your group, two minutes.**

Why can no one tickle themselves?

# Depth from Flat Images

A **depth cue** indicates distance, given an assumption that is usually true. **Motion parallax**, near things drifting faster than far ones as the observer moves, needs movement.

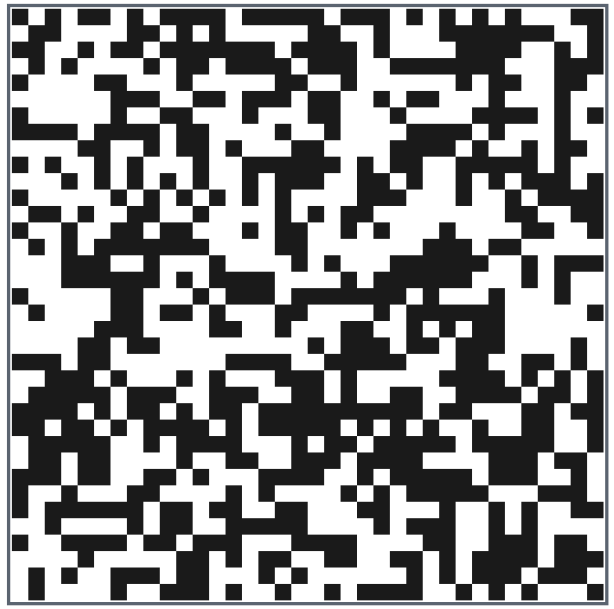


# Two Eyes, One Matching Problem

**Binocular disparity**: the mismatch between where a point falls in the two eyes. The **correspondence problem**: pairing each point with its partner first.

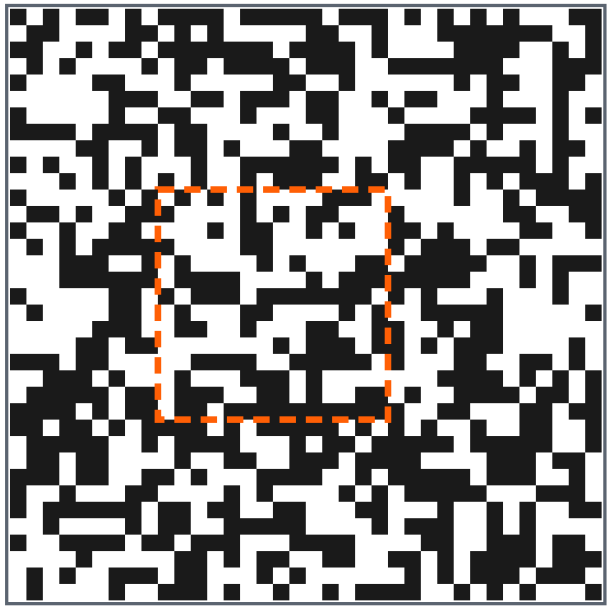
**A square that exists only in the difference between two images**

**left eye**



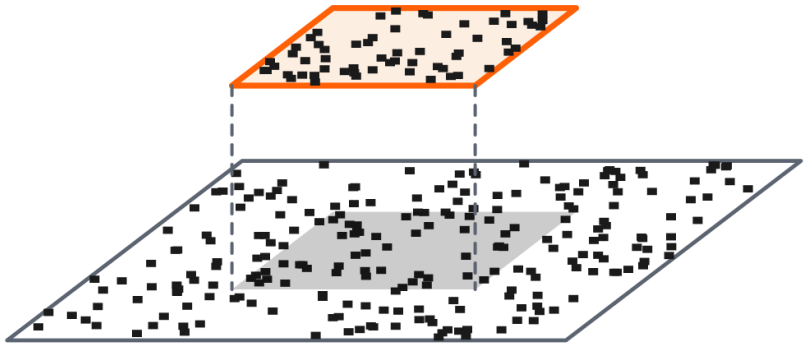
*neither picture alone contains a square*

**right eye**



*dashed outline: the shifted patch, marked here only, not in the picture*

**seen with both eyes**



*a square floats in front of the background*

# Color from Comparisons

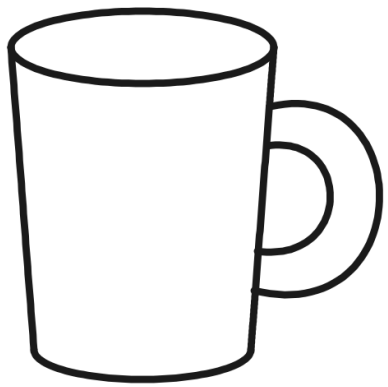
Color begins with a comparison between kinds of receptor. Constancy needs more than one surface's light, and neither proposed procedure is settled.

| What has to be recovered | What is compared | Where it stands |
|---|---|---|
| Wavelength | Receptors tuned to different wavelengths: three kinds in humans, like a camera's three filters | Needed: one receptor cannot tell dim light of one wavelength from brighter light of another |
| Constancy, by scene comparison | Each surface against the rest of the scene: if all the light turns reddish, every surface sends more red | Proposed; an early ratio version did not work everywhere; white balance is a cruder version |
| Constancy, by learning | The light from a surface against what similar light, in similar surroundings, has usually signified | Proposed |

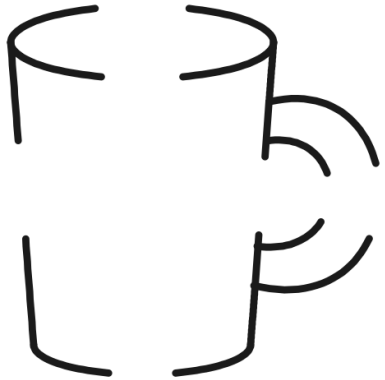
# Recognizing Objects: Parts or Views?

Brains achieve tolerance, not perfect invariance. Two accounts of how, now usually seen as ends of a continuum of tasks.

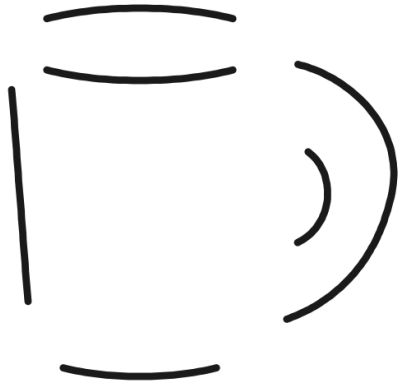
**The same amount of contour removed in two ways**



**complete**



**recoverable**



**nonrecoverable**

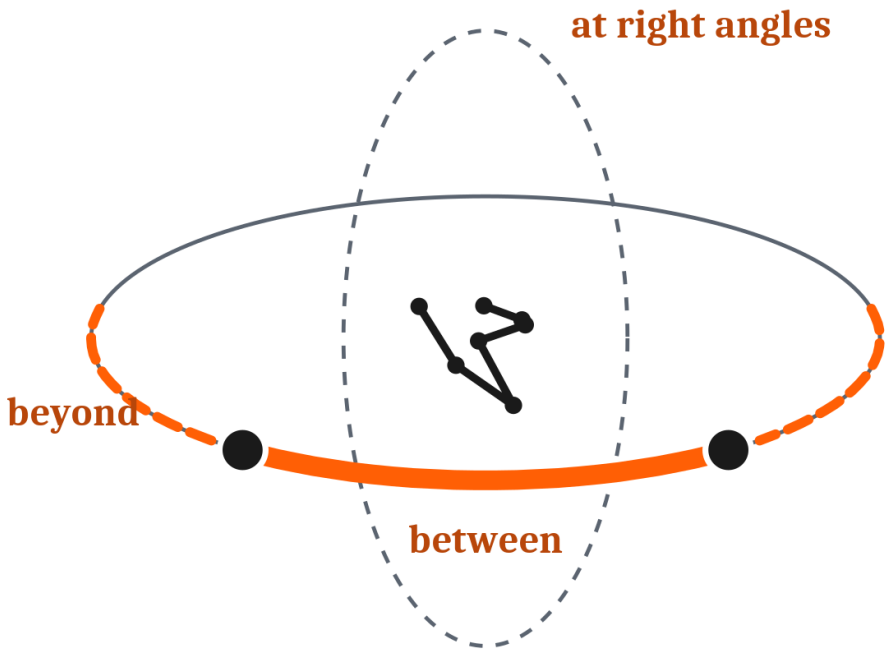
*after Biederman (1987); the gaps total about the same length in both*

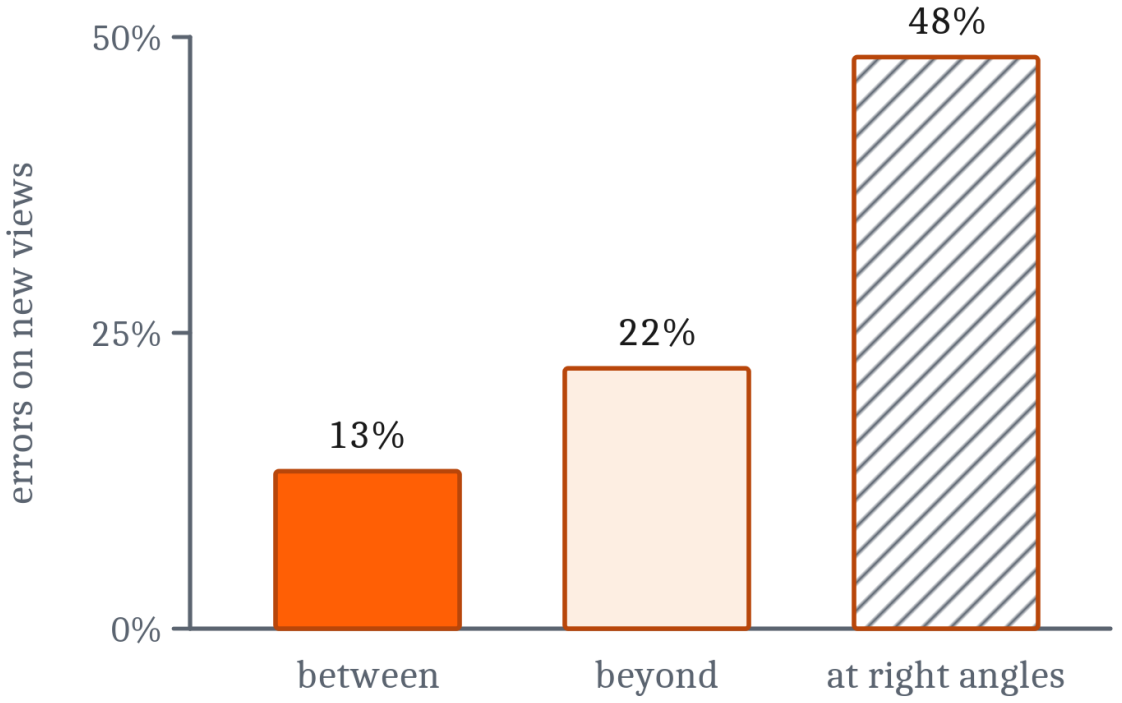
# Recognizing Objects: Parts or Views?

Brains achieve tolerance, not perfect invariance. Two accounts of how, now usually seen as ends of a continuum of tasks.

**Recognizing a new view of a novel object**

two learned views (black dots) on one circle around the object



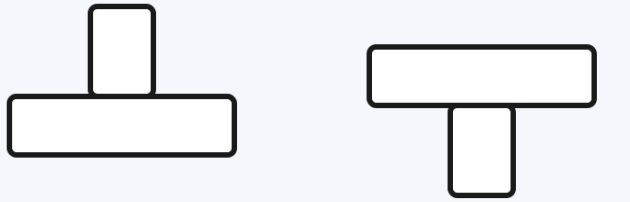




*four observers; data from Bülthoff and Edelman (1992)*

# What Networks Share with Us, and What They Do Not

Networks fail where the parts account predicts. Training on altered copies gave tolerance to those alterations, and very little beyond.

## One part moved



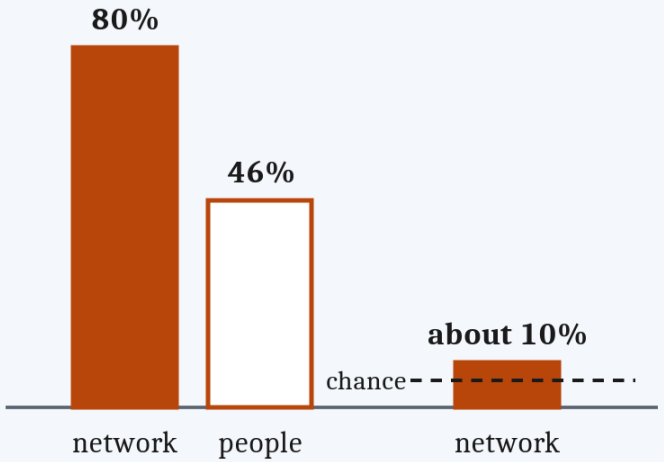
learned block moved below

**People: rarely confused**

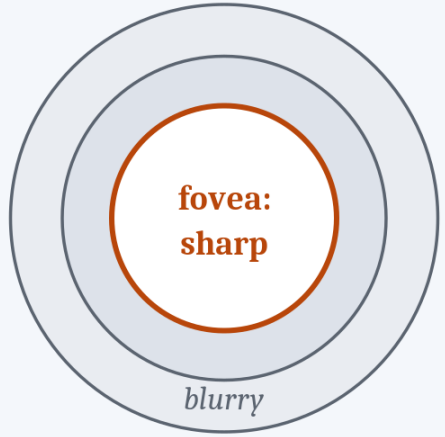
**Networks: confused easily**

*the parts account's prediction*

## Training on altered copies




## What each learns from




An eye: one small sharp window that keeps jumping.
A network: sharp photographs.

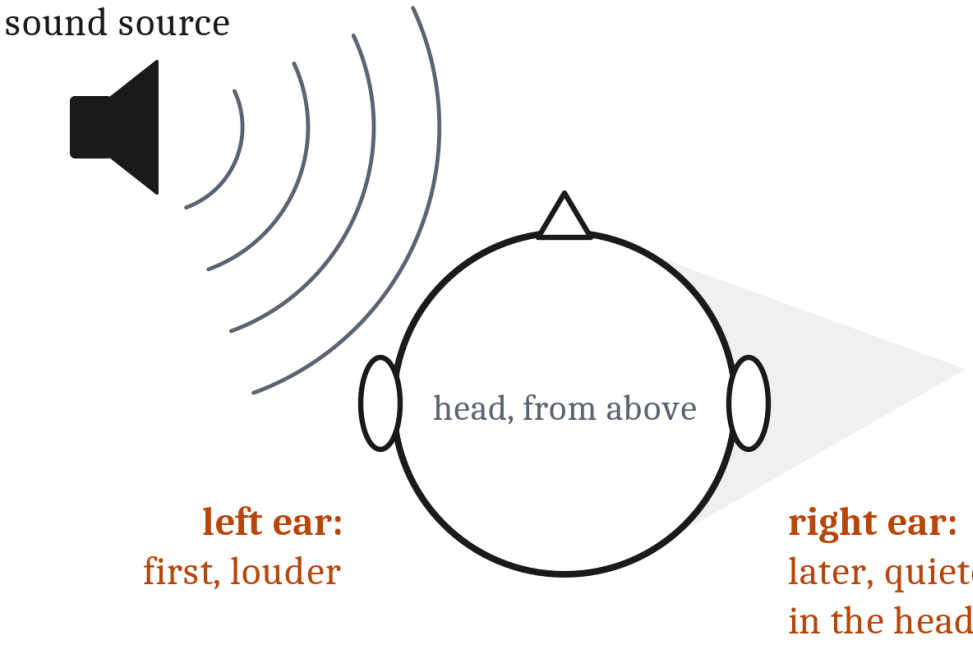
A model that matches people is not thereby shown to work like a brain.

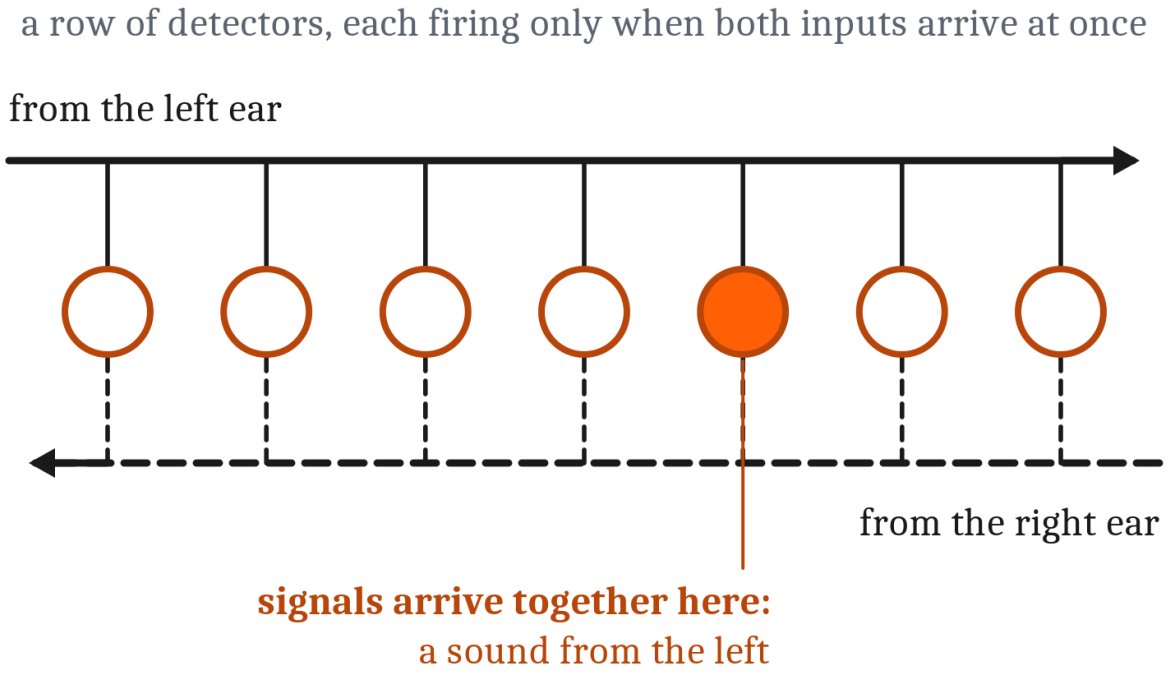
**The strongest tests change one thing, and ask whether the model fails where people fail.**

# Where a Sound Comes From

The **interaural time difference** (the near ear hears first) and **interaural level difference** (the head shadows the far ear). A *delay line* turns a time difference into a place.

**Turning a difference in time into a place**

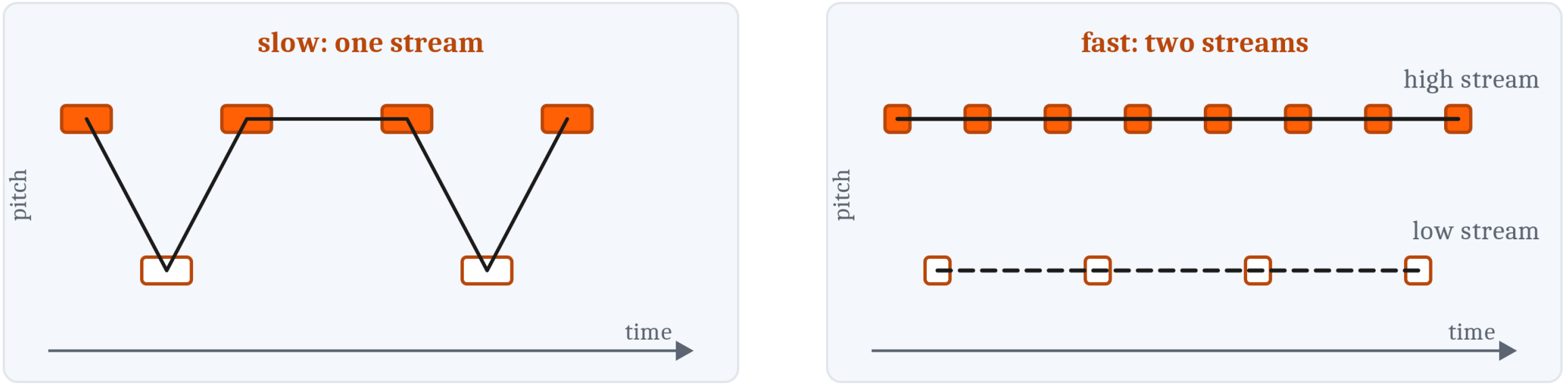


# Same Source? Same Cause?

**Auditory scene analysis** groups one mixed wave into its sources. **Cue combination** weights each source of evidence by its reliability.

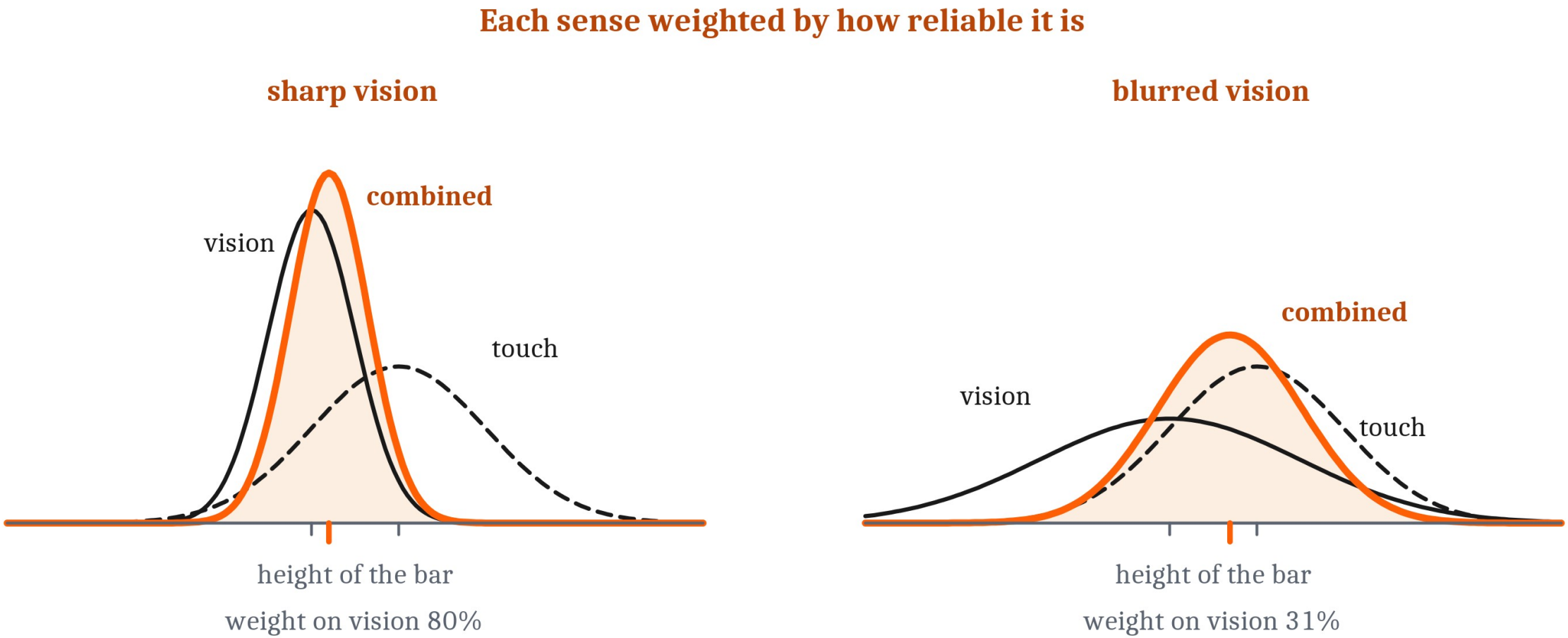
**Two tones heard as one stream, or as two**




*filled bars, high tone; open bars, low tone*

# Same Source? Same Cause?

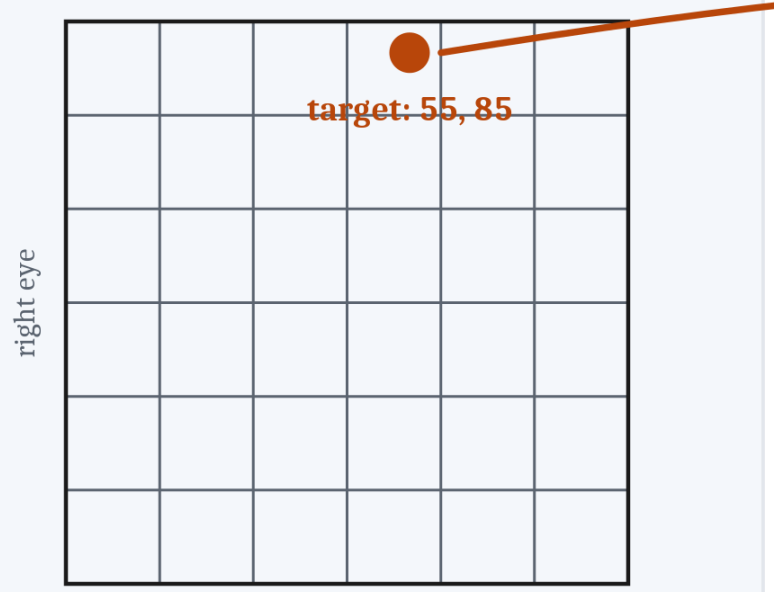
**Auditory scene analysis** groups one mixed wave into its sources. **Cue combination** weights each source of evidence by its reliability.




*solid, vision; dashed, touch; thick and shaded, the combined estimate*

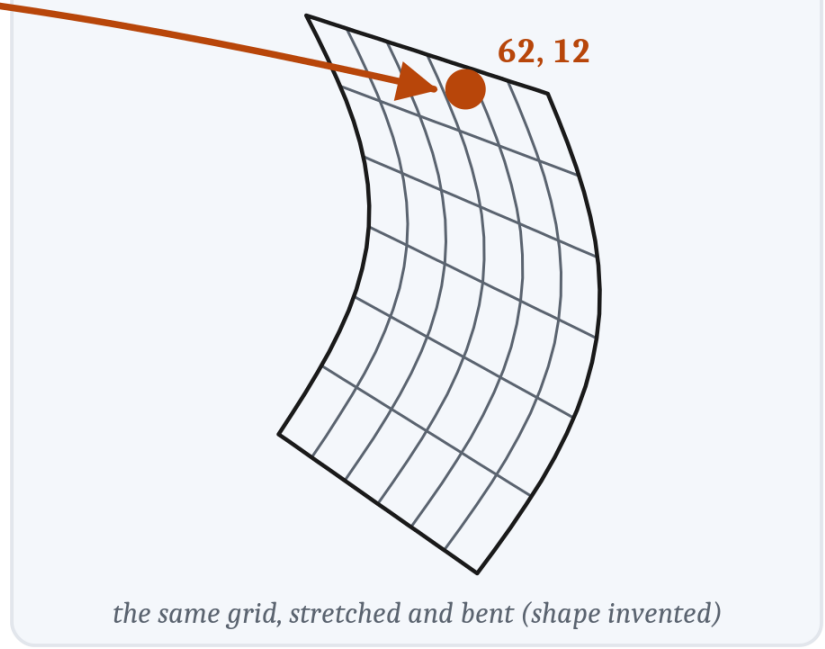
# From Eye to Arm

Acting needs a mapping between coordinate systems that share no numbers, and the mapping changes every time the eyes move.




Copying the eye numbers across would send the arm nowhere near the target.

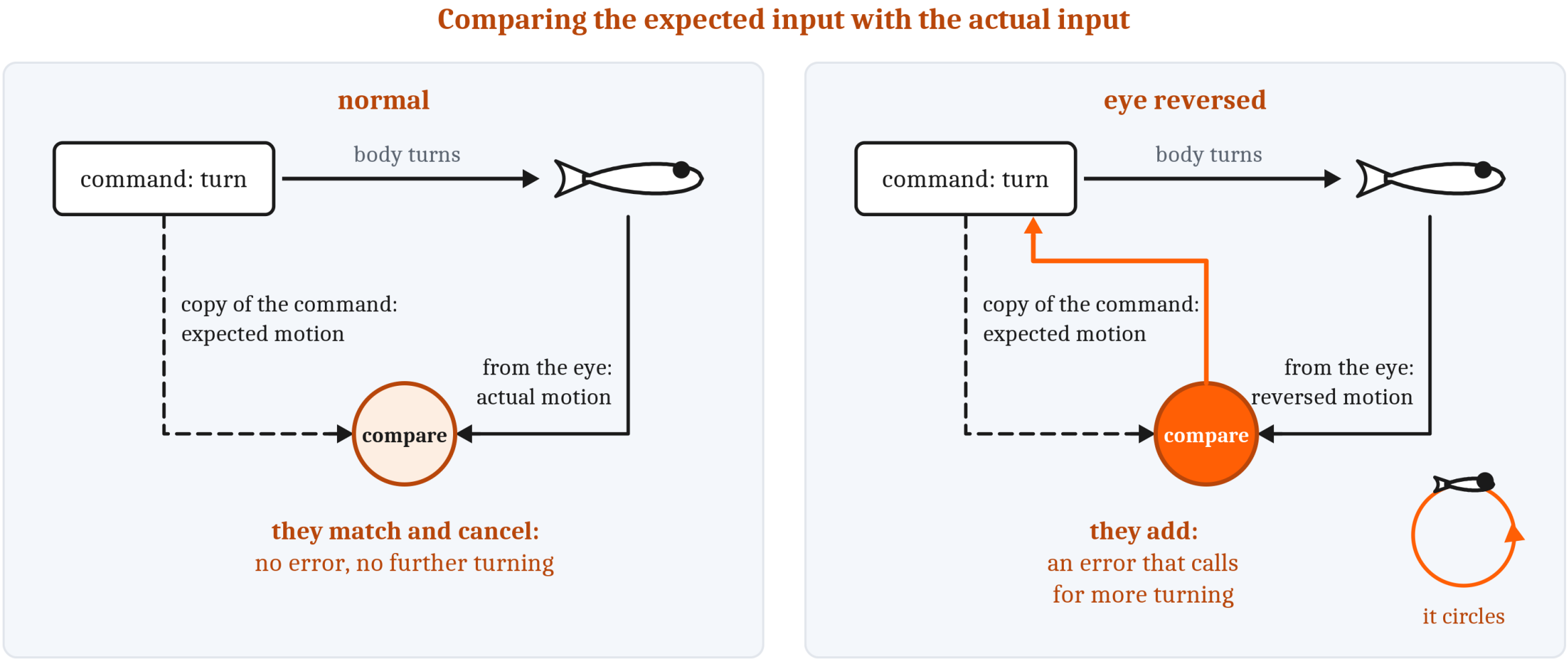
A table could store every answer. A network learns it with far fewer values, and can relearn it.




After a saccade, one spot on the retina stands for a new direction, so the mapping needs a running **record of where the eyes point. Monkeys keep one.**

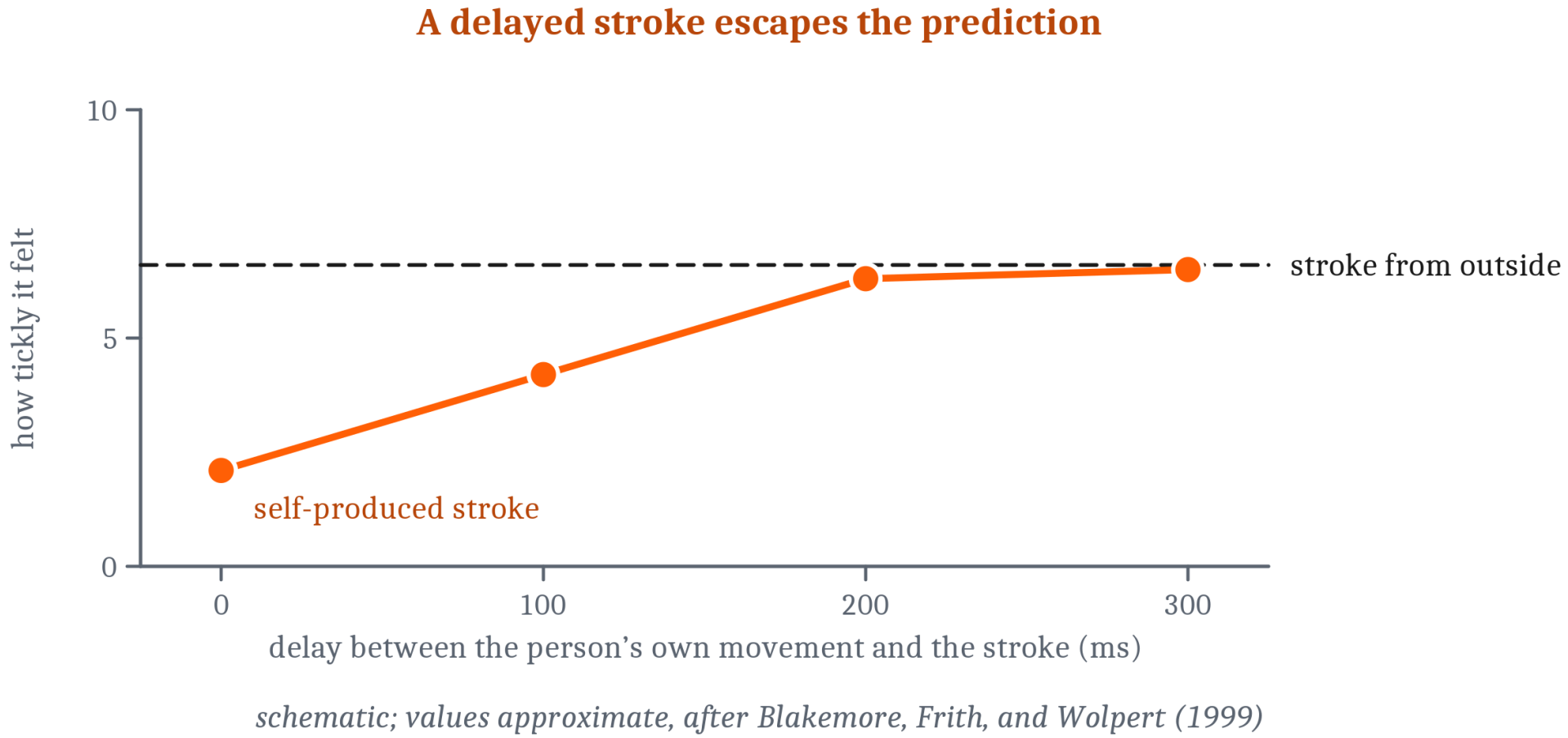
# A Copy of the Command

Reverse an eye, and expected and actual motion add instead of canceling. An **efference copy** is a copy of a command kept for the comparison; **corollary discharge** is now used almost interchangeably.



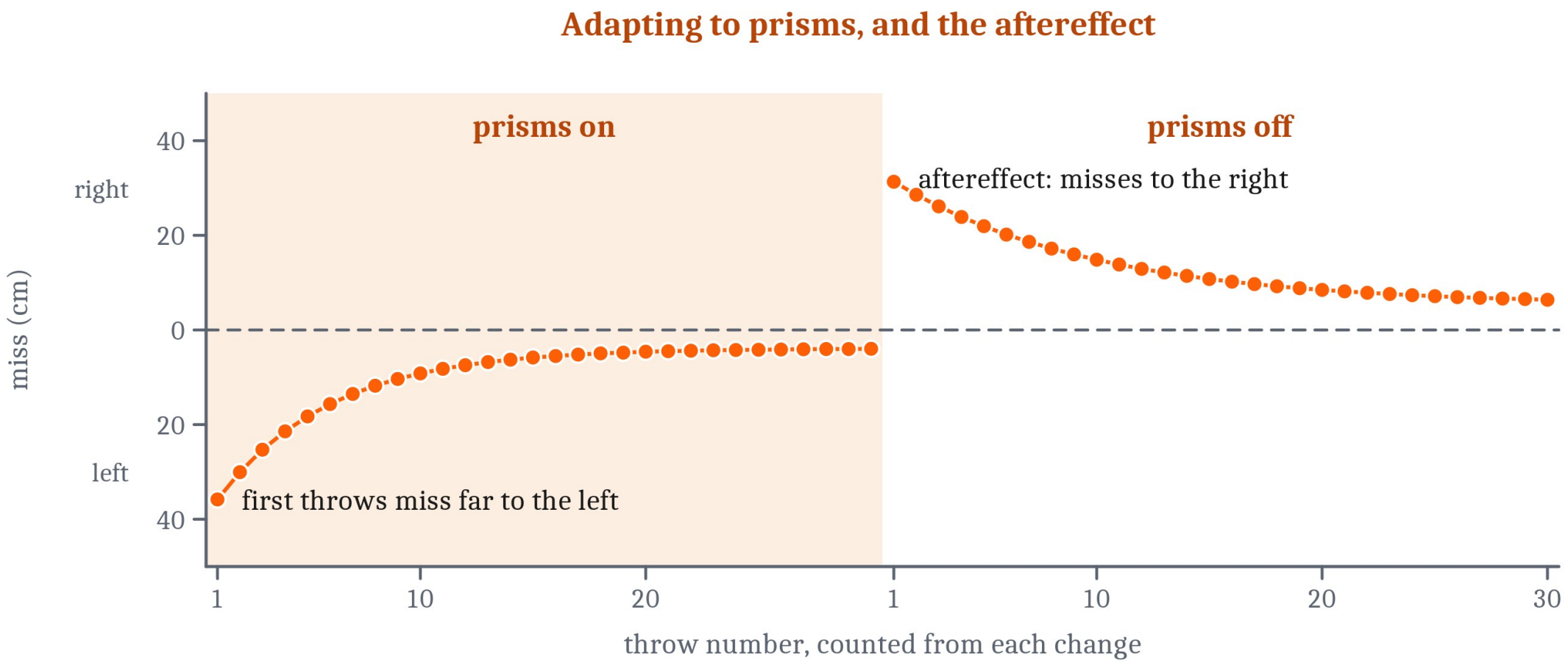
# Predicting One's Own Sensations

A **forward model** predicts what a command will do to body and senses. An **inverse model** runs from a wanted outcome to the command.



# Learning the Model from Mismatch

The model is learned from prediction error. The **aftereffect**, a miss the other way once the distortion is removed, shows that a stored mapping changed.




*fitted curves for one thrower, redrawn after Martin et al. (1996)*

# Is a World Model a Forward Model?

**In your group, four minutes.**

Several AI companies now announce “world models”. Some are systems trained on video to predict what a scene will show next, given an action.

Is a system like that a forward model? What would it need to predict, and from what?

If one matched people well on some test, what test would give the strongest evidence about whether it works the way a brain does?

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